

FEET AND CALVES

The muscles that move the ankle and toes are located primarily in the lower leg and can be described as muscles with tendons as long as or longer than the muscle. The predominant tendon is the Achilles tendon. Three muscles (gastrocnemius, plantaris, soleus) share this tendon, and the three collectively are called the triceps surae. The triceps surae muscles are the prime plantar flexors and are assisted by the popliteus and posterior tibialis as well as flexor digitorum longus and flexor hallucis longus, which also move the toes. Another group of three muscles (peroneus longus, peroneus brevis, peroneus tertius) is located on the outer (lateral) side of the calf; these muscles are used in pulling the inner ankles toward the floor. The anterior calf muscles (anterior tibialis, extensor hallucis longus, extensor digitorum longus) not only dorsiflex the ankle but also move the foot and toes. The muscles on the dorsal (top) side of the foot (extensor digitorum brevis, dorsal interosseous, extensor hallucis brevis) extend the toes. The muscles on the plantar (sole) side of the foot (flexor digitorum brevis, quadratus plantae, flexor hallucis brevis, flexor digiti minimi, abductor hallucis, abductor digiti minimi, plantar interosseous, lumbricales) are used to flex and spread the toes. Figures showing these muscles as well as a chart showing specific movements for each muscle are located at the end of the chapter (pages 138-140).

In normal daily activities, the muscles of the foot and lower leg are used more extensively than any other muscles in the body. Though the musculature of the lower leg is substantially smaller than that of the upper leg, it essentially supports the whole body and receives the heaviest load during walking or standing. As a result, many people have minor aches and pains in these muscles. Thus, at the end of the day people are ready to sit down and let these muscles rest. Stretching and strengthening these smaller muscle groups can alleviate some of the daily fatigue and pain. Stretching can also improve flexibility and stamina. These improvements enable the muscle groups to work harder and longer throughout the day.

Soreness, tightness, cramping, restlessness, and weakness in the arch of the foot and calf muscles are common complaints among people. These problems often result from the continuous and heavy load put on the muscle. Chronic use of these muscles can also increase muscle tightness and soreness. Tightness then leads to conditions such as tendinitis and shinsplints; tendinitis of the Achilles tendon is quite common. Tendinitis is associated with overuse and tightness of the gastrocnemius and soleus muscles. Shinsplints are associated with inflammation of the frontal compartment of the lower-leg muscles—anterior tibialis and, in some cases, soleus and flexor digitorum longus. These conditions can become excruciating if not treated in the early stages. A variety of stretching and strengthening exercises in those muscle groups will, in most cases, improve these conditions (lessen the severity) and help prevent future episodes from occurring. People often have delayed-onset muscle soreness, or DOMS, after participating in unusual or unfamiliar activities. Calf muscles tend to be affected by delayed-onset muscle soreness more often than any other muscle group in the body. Light stretching exercises help to improve this condition and relieve some of the pain associated with it.

All of the instructions and figures in this chapter are given for the right side of the body. Similar but opposite procedures would be used for the left side of the body.

Seated Toe Extensor Stretch

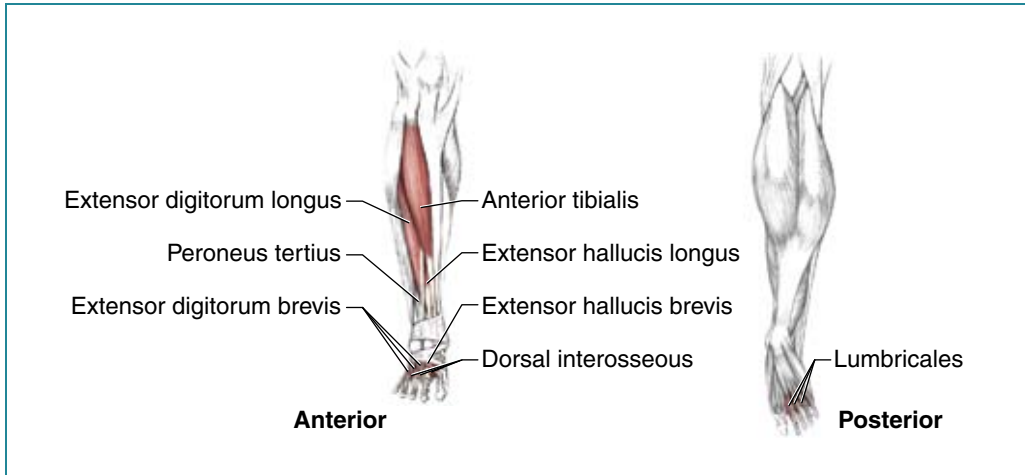


Technique

While sitting on a chair with the left foot on the floor, raise the right ankle and place it on top of the left knee.

While bracing the right ankle with the right hand, place the fingers of the left hand on the tops of the right toes.

Pull the tips of the toes toward the sole of the foot.



Muscles Stretched

Most-stretched muscles: Right extensor digitorum longus, right extensor digitorum brevis, right extensor hallucis longus, right extensor hallucis brevis, right anterior tibialis.

Lesser-stretched muscles: Right peroneus tertius, right dorsal interosseous, right lumbricales.

Commentary

Hold the ankle firmly in order to keep it and the foot stable. You will feel the stretch on the top of the foot area (dorsal side). If grasping and pulling on the tips of the toes cause too much pain, apply the pressure at the ball of the foot.

Seated Toe Extensor and Foot Everter Stretch

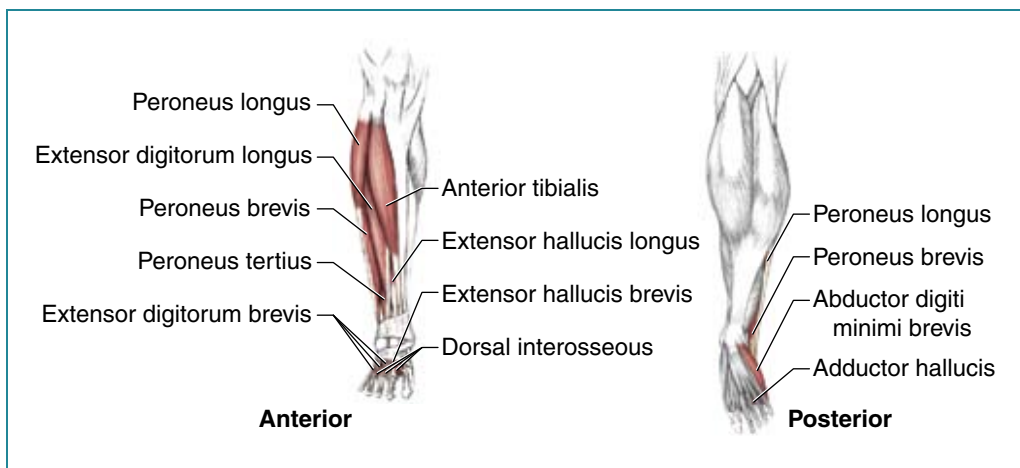


Technique

While sitting on a chair with the left foot on the floor, raise the right ankle and place it on top of the left knee.

While bracing the right ankle with the right hand, place the thumb of the left hand along the ball of the right foot and place the fingers of the left hand across the top of the foot with the fingers perpendicular to the toes.

Use the left hand to pull (or twist) the sole of the foot upward. At the same time, bend the toes toward the sole of the foot.



Muscles Stretched

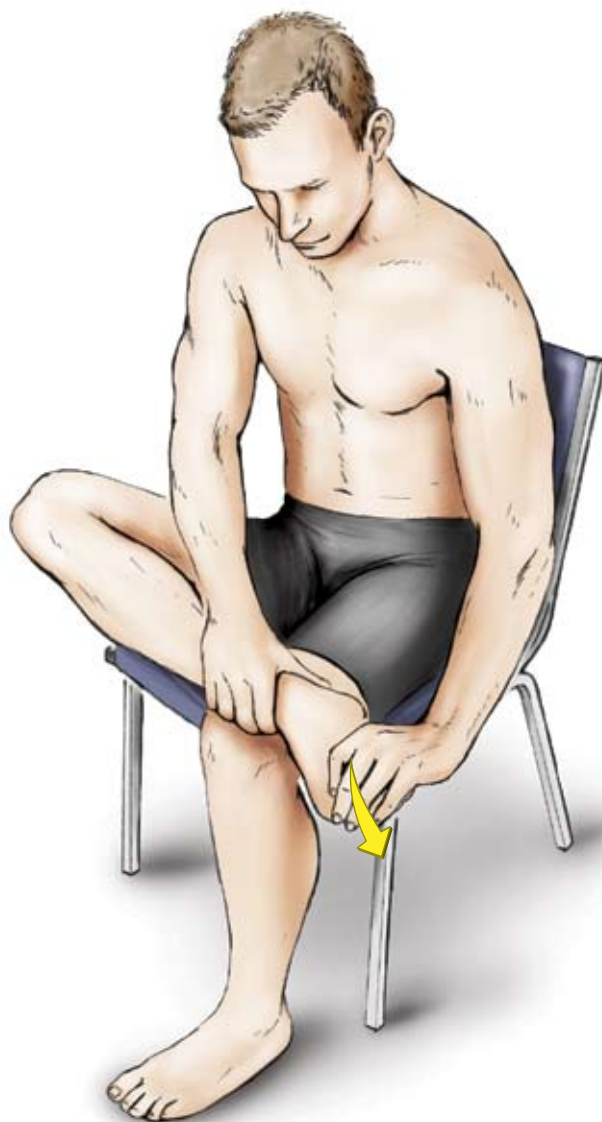
Most-stretched muscles: Right extensor digitorum longus, right extensor digitorum brevis, right extensor hallucis longus, right extensor hallucis brevis, right dorsal interosseous, right abductor digiti minimi brevis, right adductor hallucis, right peroneus longus, right peroneus brevis, right peroneus tertius.

Lesser-stretched muscle: Right anterior tibialis.

Commentary

Make sure to stabilize the foot and ankle with a firm hold. Grasping the ends of the toes and pulling them upward (while keeping the toes in the flexed position) can produce a more effective stretch. You will feel the stretch on the lateral side of the foot (little toe side) and ankle area—abductor digiti minimi, extensor digitorum brevis, and extensor hallucis brevis muscles.

Seated Toe Extensor and Foot Inverter Stretch

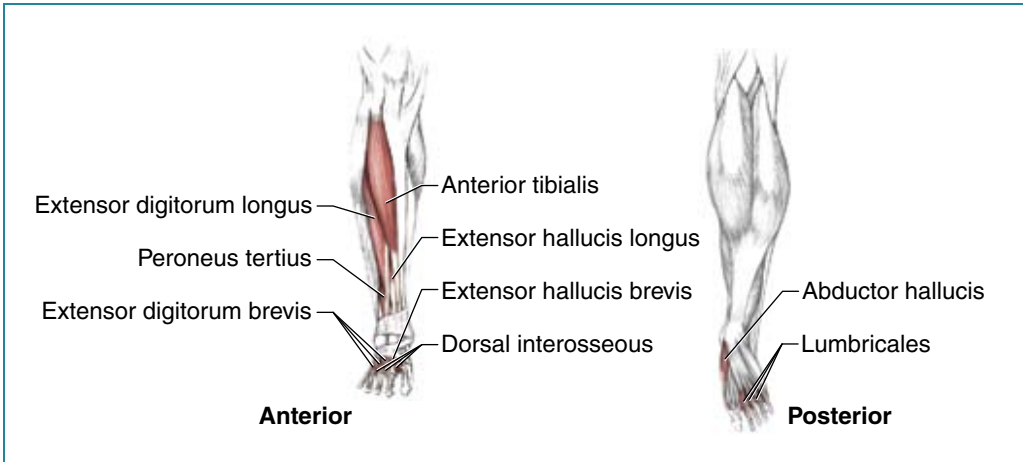


Technique

While sitting on a chair with left foot on the floor, raise the right ankle and place it on top of the left knee.

While bracing the right ankle with the right hand, place the thumb of the left hand along the ball of the right foot and the fingers of the left hand across the top of the foot with the fingers perpendicular to the toes.

Use the left hand to push (or twist) the sole of the foot down toward the floor. At the same time, bend the toes toward the sole of the foot.



Muscles Stretched

Most-stretched muscles: Right abductor hallucis, right extensor hallucis longus, right extensor hallucis brevis, right anterior tibialis.

Lesser-stretched muscles: Right extensor digitorum longus, right extensor digitorum brevis, right peroneus tertius, right lumbricales, right dorsal interosseous.

Commentary

Make sure to stabilize the foot and ankle with a firm hold. Grasping the ends of the toes and pressing them firmly downward can produce a greater stretch. You will feel the stretch on the medial side of the foot (big toe side).

Seated Toe Flexor Stretch

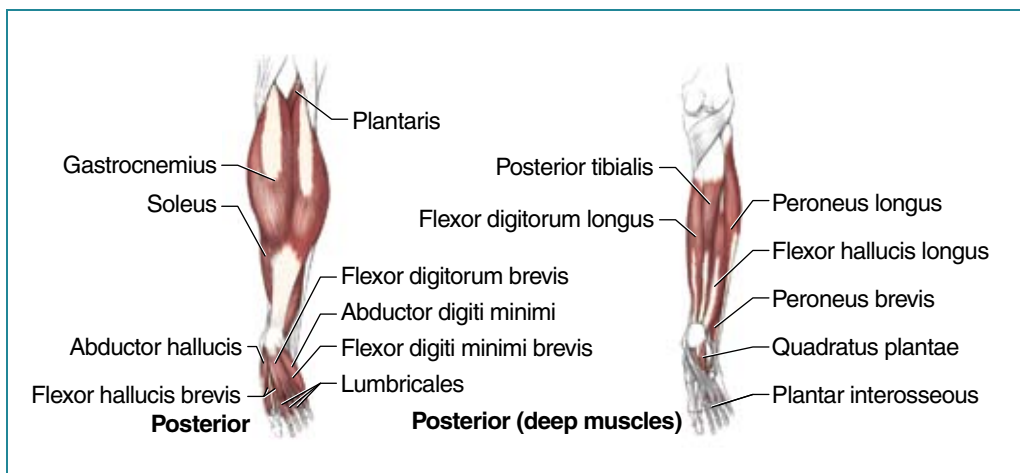


Technique

While sitting on a chair with the left foot on the floor, raise the right ankle and place it on top of the left knee.

Brace the right ankle with the right hand, and place the fingers of the left hand along the bottoms of the toes of the right foot with the fingers pointing in the same direction as the toes.

Use the fingers of the left hand to push the toes of the right foot toward the right knee.



Muscles Stretched

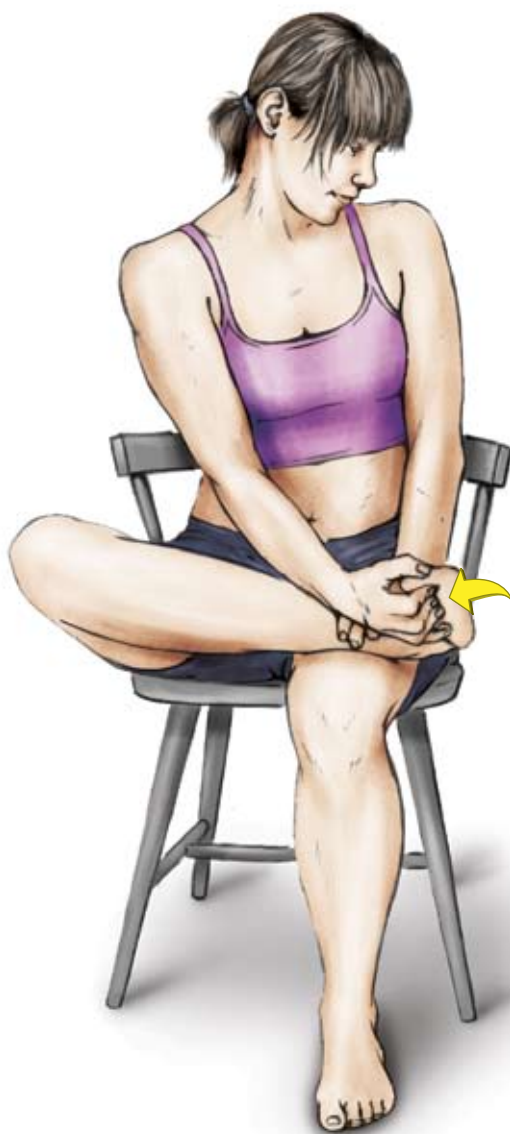
Most-stretched muscles: Right flexor digitorum brevis, right quadratus plantae, right flexor digiti minimi brevis, right flexor hallucis brevis, right lumbricales, right plantar interosseous, right abductor hallucis, right abductor digiti minimi.

Lesser-stretched muscles: Right flexor digitorum longus, right flexor hallucis longus, right posterior tibialis, right peroneus longus, right peroneus brevis, right plantaris, right soleus, right gastrocnemius.

Commentary

Make sure to stabilize the foot and ankle with a firm hold. Pushing hard on the very ends of the toes with the left palm will provide a much greater stretch. You will feel the stretch on the sole (plantar side) of the foot.

Seated Toe Flexor and Foot Everter Stretch

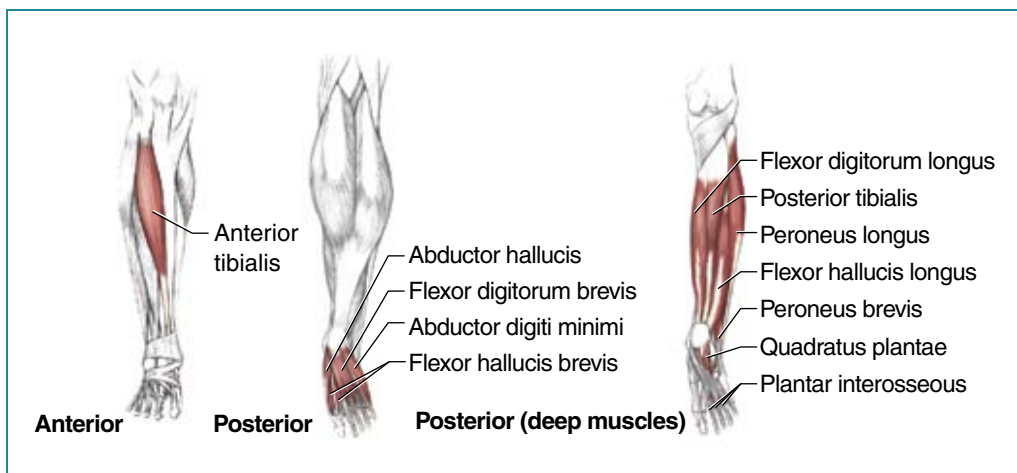


Technique

While sitting on a chair with the left foot on the floor, raise the right ankle and place it on top of the left knee.

Brace the right ankle with the left hand and place the fingers of the right hand perpendicular across the bottoms of the toes. Also place the pad of the right thumb on the ball of the right big toe.

Use the right hand to pull (or twist) the sole of the foot upward. At the same time, use the fingers of the right hand to pull the toes of the right foot up toward the top of the foot.



Muscles Stretched

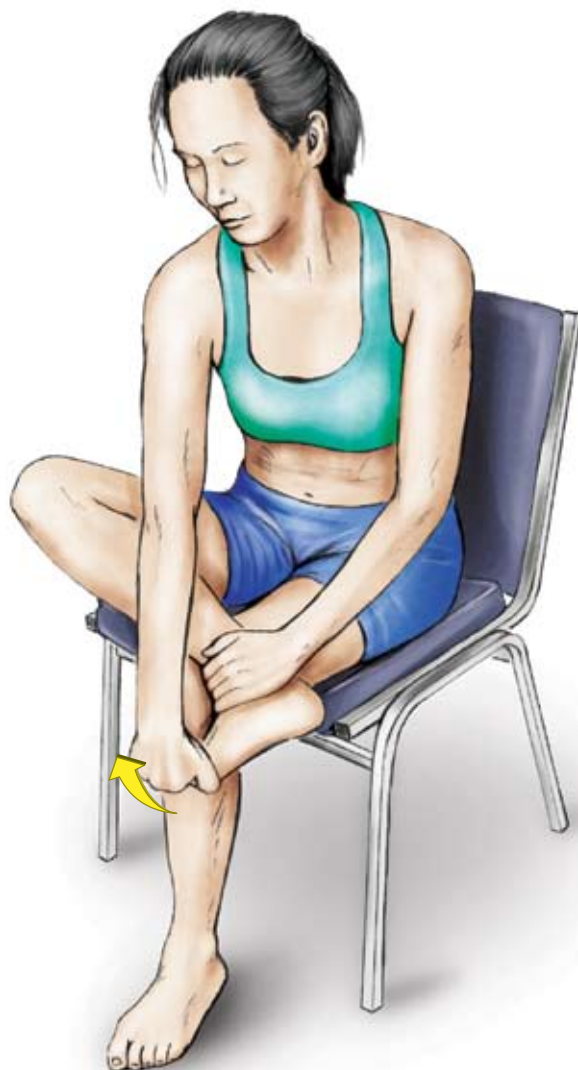
Most-stretched muscles: Right flexor digitorum brevis, right flexor hallucis brevis, right quadratus plantae, right abductor digiti minimi, right peroneus longus, right peroneus brevis, right plantar interosseous.

Lesser-stretched muscles: Right anterior tibialis, right flexor hallucis longus, right flexor digitorum longus, right posterior tibialis, right abductor hallucis.

Commentary

Make sure to stabilize the foot and ankle with a firm hold. If you grasp the very ends of the toes and pull harder, then you will be able to stretch these muscles even farther. You will feel the stretch on the sole (plantar side) of the foot.

Seated Toe Flexor and Foot Inverter Stretch

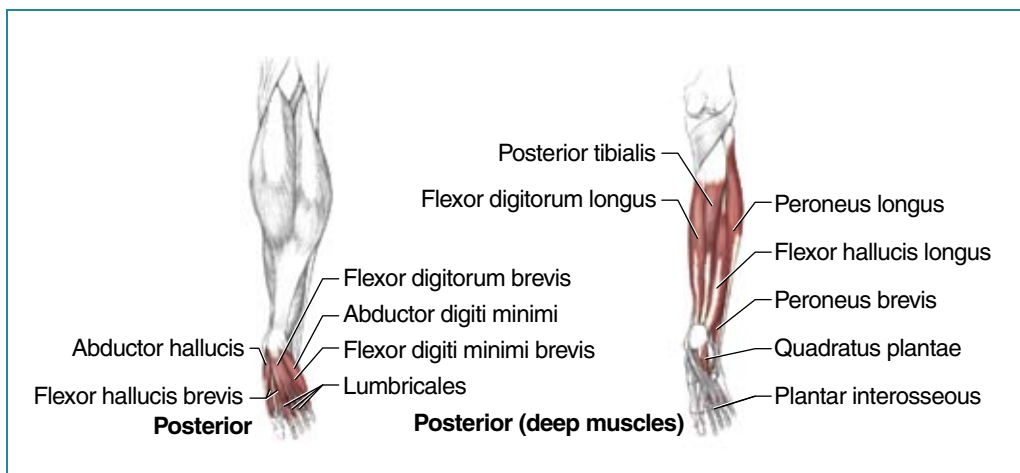


Technique

While sitting on a chair with the left foot on the floor, raise the right ankle and place it on top of the left knee.

Brace the right ankle with the left hand and place the fingers of the right hand perpendicular across the bottoms of the toes. Also place the pad of the right thumb on the ball of the right big toe.

Use the fingers of the right hand to pull the toes of the right foot up toward the top of the foot. At the same time, use the right thumb to push the sole of the right foot down toward the floor.



Muscles Stretched

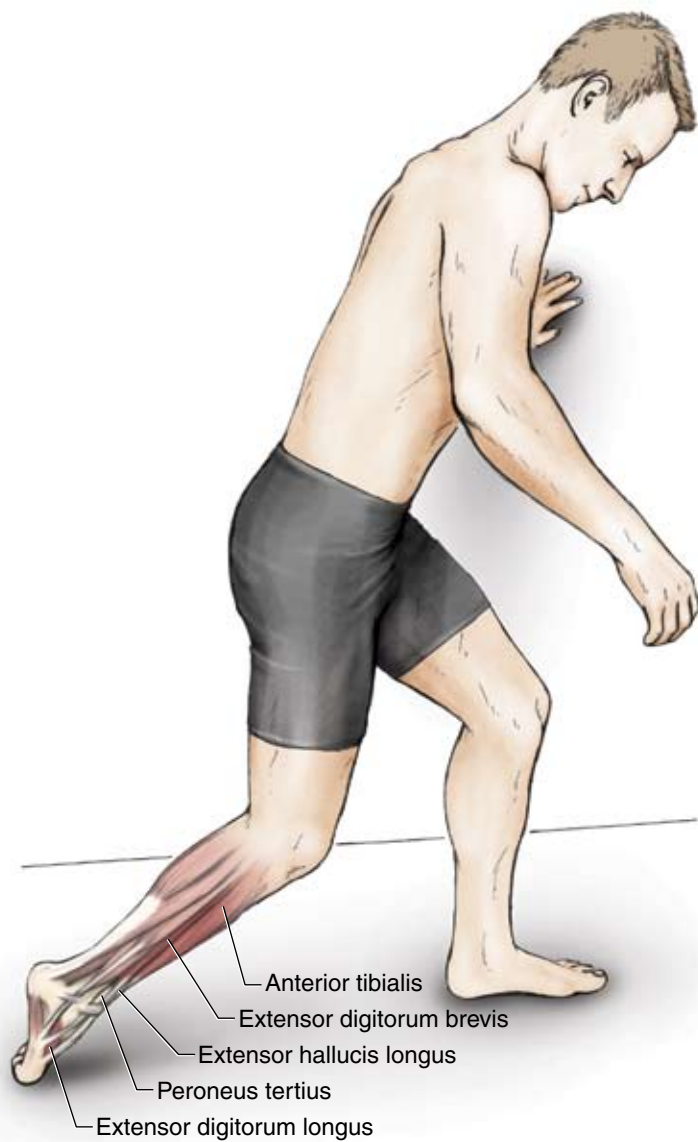
Most-stretched muscles: Right flexor digitorum brevis, right quadratus plantae, right flexor digiti minimi brevis, right flexor hallucis brevis, right lumbricales, right plantar interosseous, right abductor hallucis.

Lesser-stretched muscles: Right peroneus longus, right peroneus brevis, right abductor digiti minimi, right flexor digitorum longus, right flexor hallucis longus, right posterior tibialis.

Commentary

Make sure to stabilize the foot and ankle with a firm hold. If you grasp the very ends of the toes and pull harder, then you will be able to stretch these muscles even farther. You will feel the stretch on the sole (plantar side) of the foot—the flexor digitorum brevis, flexor hallucis brevis, flexor digiti minimi brevis, and quadratus plantae muscles.

Standing Toe Extensor Stretch



Technique

Stand upright and brace against a wall or an object for balance.

Point the right foot backward away from the body, dorsal (top) side of the toes down against the floor.

While keeping the dorsal side of the toes pressed against the floor, lean your weight onto the right leg and press the bottom of the heel down toward the floor.

Muscles Stretched

Most-stretched muscles: Right extensor digitorum brevis, right extensor hallucis brevis, right anterior tibialis, right peroneus tertius.

Lesser-stretched muscles: Right extensor hallucis longus, right extensor digitorum longus, right dorsal interosseous.

Commentary

It is more comfortable to perform this stretch on a carpet or other soft surface. Be sure not to drag the foot that is pressed to the floor. Moving the heel medially or laterally will place greater stretch on either the dorsal medial or dorsal lateral parts of the foot.

Standing Toe Flexor Stretch



Technique

Stand upright while facing the wall, 1 to 2 feet (30 to 61 cm) away.

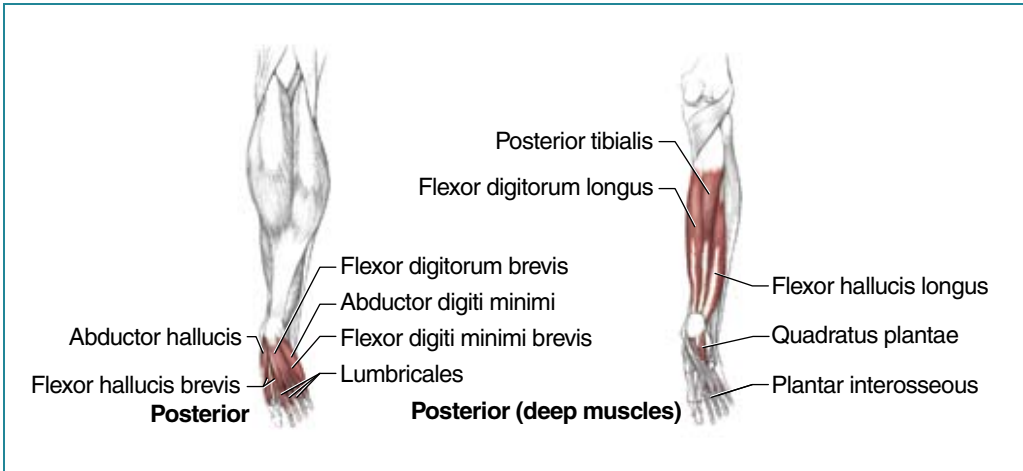
Keeping the heel of the foot on the floor, press the bottoms of the toes of the right foot up against the wall. The ball of the foot should be more than half an inch (more than 2 cm) above the floor.

Lean forward and slide the ball of the foot slowly down, keeping the toes pressed against the wall.

Muscles Stretched

Most-stretched muscles: Right flexor digitorum brevis, right quadratus plantae, right flexor digiti minimi brevis, right flexor hallucis brevis, right lumbricales, right plantar interosseous, right abductor hallucis, right abductor digiti minimi.

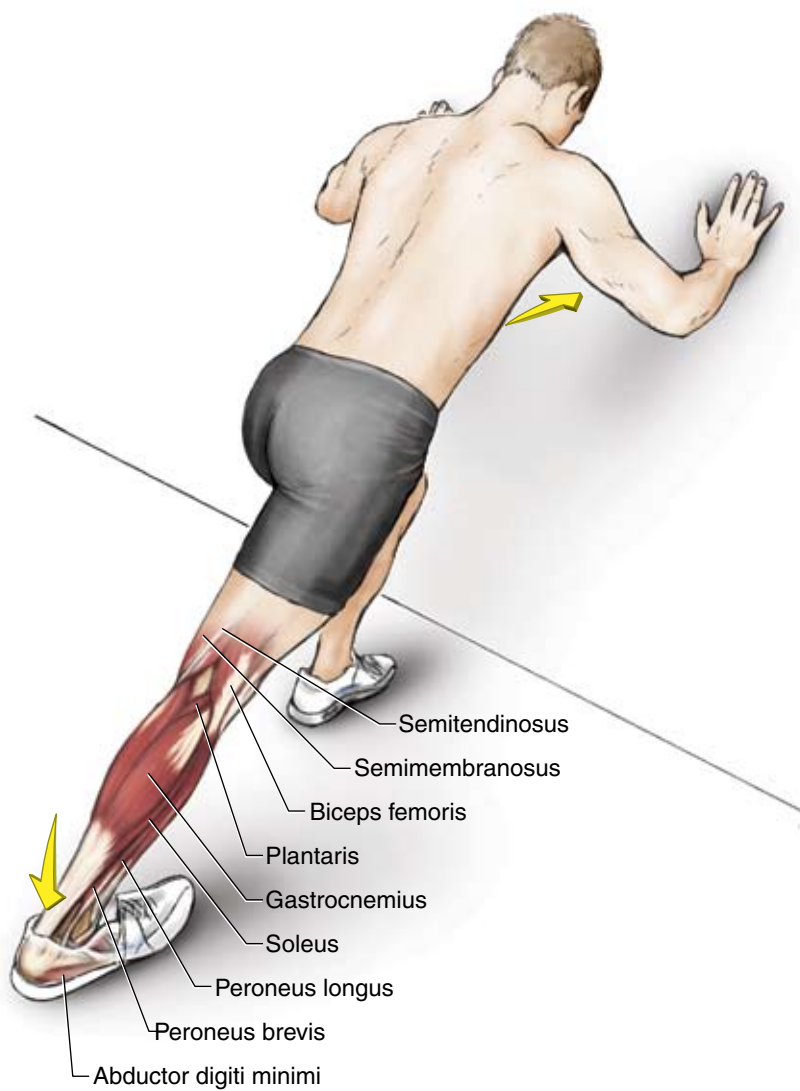
Lesser-stretched muscles: Right flexor digitorum longus, right flexor hallucis longus, right posterior tibialis.



Commentary

Make sure that the ball of the foot is parallel to the floor. This ensures that all of the toes are stretched equally. Also, slide the ball of the foot down slowly; otherwise, overstretching could happen. Bending the right knee slightly and moving the knee forward toward the wall will incorporate the calf muscles in the stretch.

Single Plantar Flexor Stretch



Technique

Stand facing a wall 2 feet (61 cm) away.

Brace your hands against the wall.

Keeping the left foot in place, place the right foot 1 to 2 feet (30 to 61 cm) behind the left foot. The left foot is 1 to 2 feet away and the right foot is 2 to 4 feet (61 to 122 cm) away from the wall.

Keeping the right heel on the floor, lean your chest toward the wall. You can bend the left knee slightly to facilitate moving the chest up against the wall.

Muscles Stretched

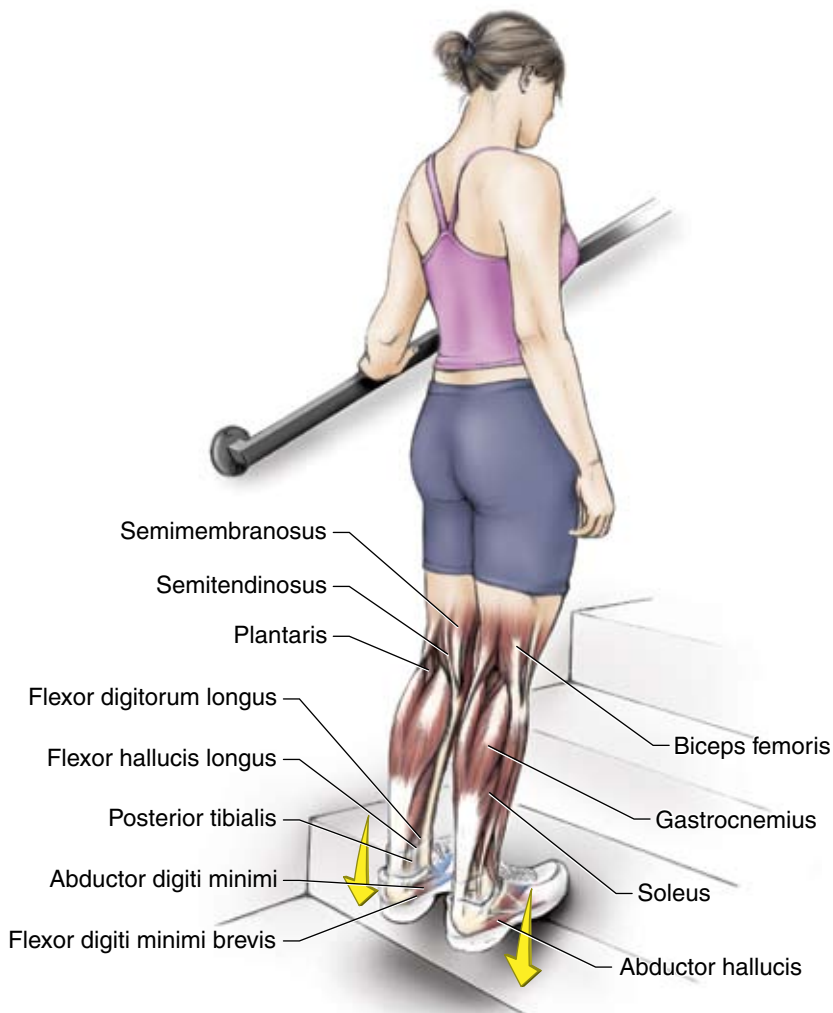
Most-stretched muscles: Right gastrocnemius, right soleus, right plantaris, right popliteus, right flexor digitorum longus, right flexor hallucis longus, right posterior tibialis.

Lesser-stretched muscles: Right peroneus longus, right peroneus brevis, right flexor digitorum brevis, right quadratus plantae, right flexor digiti minimi brevis, right flexor hallucis brevis, right abductor digiti minimi, right abductor hallucis, right popliteus, right semitendinosus, right semimembranosus, right biceps femoris.

Commentary

As the chest gets closer to the wall, bending the knee slightly will realign the tibia and increase the distance between the muscle attachment points. This will increase the stretch on the posterior tibialis, flexor hallucis longus, and flexor digitorum longus muscles while at the same time reducing the stretch on the hamstring muscles.

Double Plantar Flexor Stretch



Technique

Stand upright on the edge of a stair or beam, with both heels unsupported out past the edge.

Keep the right and left knees straight, and hold on to a support with at least one hand.

Lower both heels down as far as possible.

Muscles Stretched

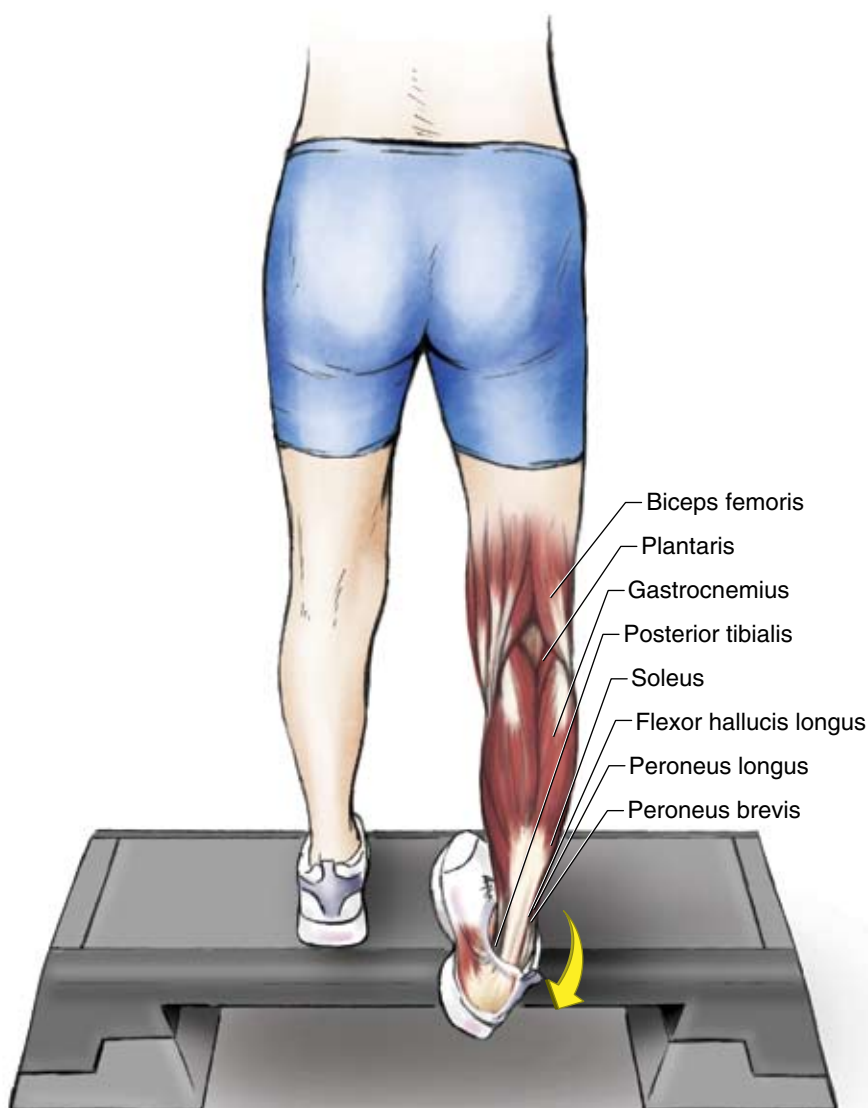
Most-stretched muscles: Gastrocnemius, soleus, plantaris, popliteus, flexor digitorum longus, flexor digitorum brevis, flexor hallucis longus, flexor hallucis brevis, posterior tibialis, quadratus plantae, flexor digiti minimi brevis, abductor digiti minimi, abductor hallucis.

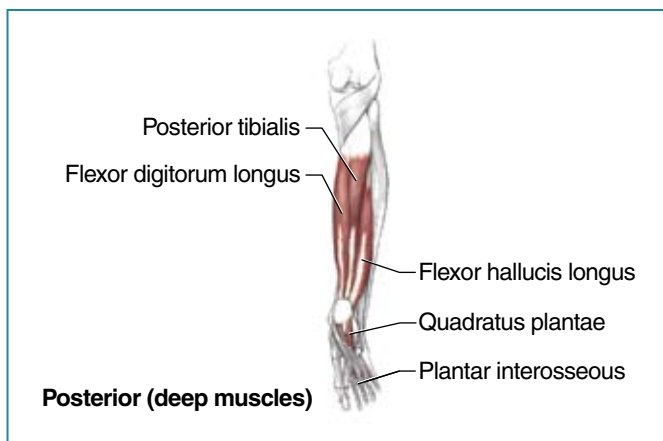
Lesser-stretched muscles: Semitendinosus, semimembranosus, biceps femoris.

Commentary

It is more comfortable to do this stretch while wearing shoes. Always support the body—an unsupported body could cause the muscles to contract and not stretch. You will increase the extent of the stretch if you work one leg at a time. After the heels reach their lowest point, you can apply more stretch by bending the knees slightly. This will stretch the posterior tibialis, flexor hallucis longus, and flexor digitorum longus muscles; at the same time it will reduce the stretch on the hamstring muscles. Placing the ball of the foot on the edge of the stairs or beam will increase the stretch on the origin (top part) of these muscle groups. Placing the midsection of the foot on the edge of the stairs or beam increases the stretch on the lower portion of these muscles.

Plantar Flexor and Foot Everter Stretch





Technique

Stand upright on the edge of a stair or beam, with the midsection of the right foot on the edge.

Place the foot in an inverted position (stand on the lateral [outer] side of the foot).

Keep the right knee straight and the left knee slightly bent.

Hold on to a support with at least one hand.

Keeping the foot inverted, lower the right heel as far as possible.

Muscles Stretched

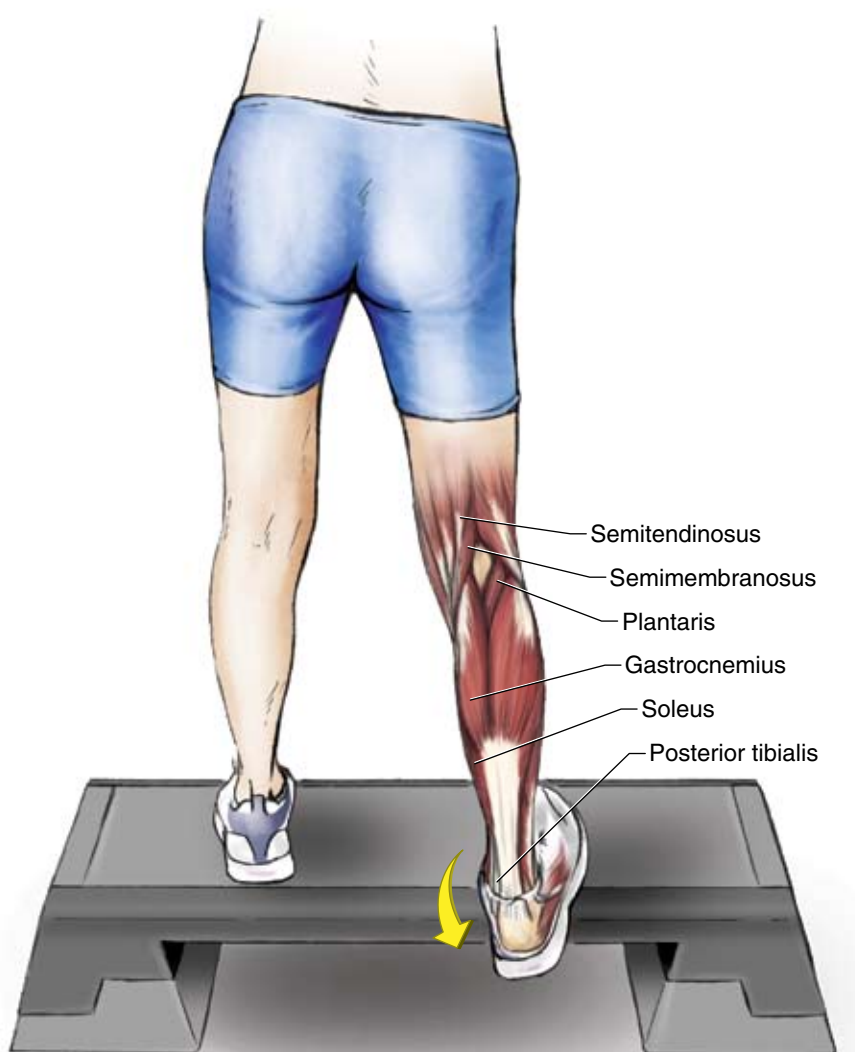
Most-stretched muscles: Right peroneus longus, right peroneus brevis, right peroneus tertius, right abductor digiti minimi, lateral side of right soleus and right gastrocnemius, right flexor hallucis longus, right posterior tibialis.

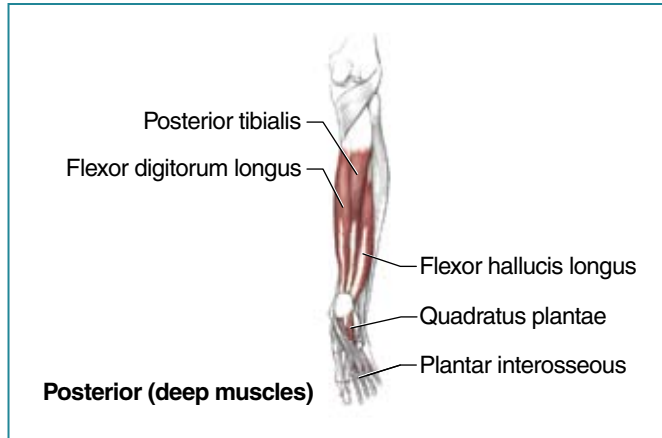
Lesser-stretched muscles: Right popliteus, right plantaris, medial head of right gastrocnemius, right biceps femoris, right flexor digitorum brevis, right quadratus plantae, right flexor digiti minimi brevis, right flexor hallucis brevis.

Commentary

It is more comfortable to do this stretch while wearing shoes. This is an excellent stretch for the peroneus longus and brevis and the abductor digiti minimi muscles, which are located at the lateral (outer) side of the lower leg and the foot. Be extra careful when placing the foot in an inverted position, and make sure to progress slowly through this stretching exercise. After the right heel reaches the floor or the lowest point, you can increase the stretch by bending the right knee slightly. This removes any stretch on hamstring muscles, but it stretches the calf muscles further.

Plantar Flexor and Foot Inverter Stretch





Technique

Stand upright on the edge of a stair or beam, with the midsection of the right foot on the edge.

Place the foot in an everted position (stand on the medial [inner] side of the foot).

Bend the right knee slightly toward the midsection of the body (inside direction), with the left knee slightly bent.

Hold on to a support with at least one hand.

While keeping the foot everted, lower the right heel as far as possible.

Muscles Stretched

Most-stretched muscles: Right flexor digitorum longus, right abductor hallucis, medial side of right soleus, right posterior tibialis, right plantaris.

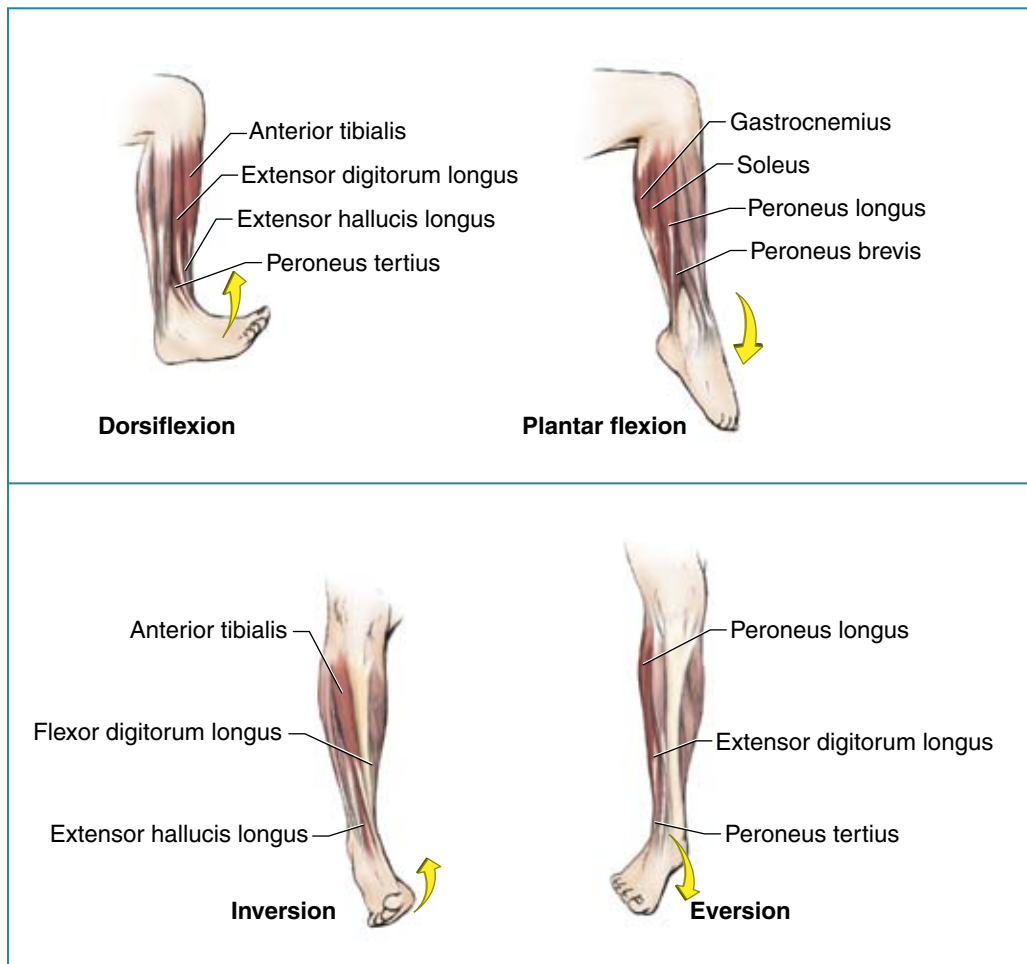
Lesser-stretched muscles: Right flexor digitorum brevis, right quadratus plantae, right flexor hallucis brevis, right flexor digiti minimi brevis, right medial gastrocnemius, right semitendinosus, right semimembranosus.

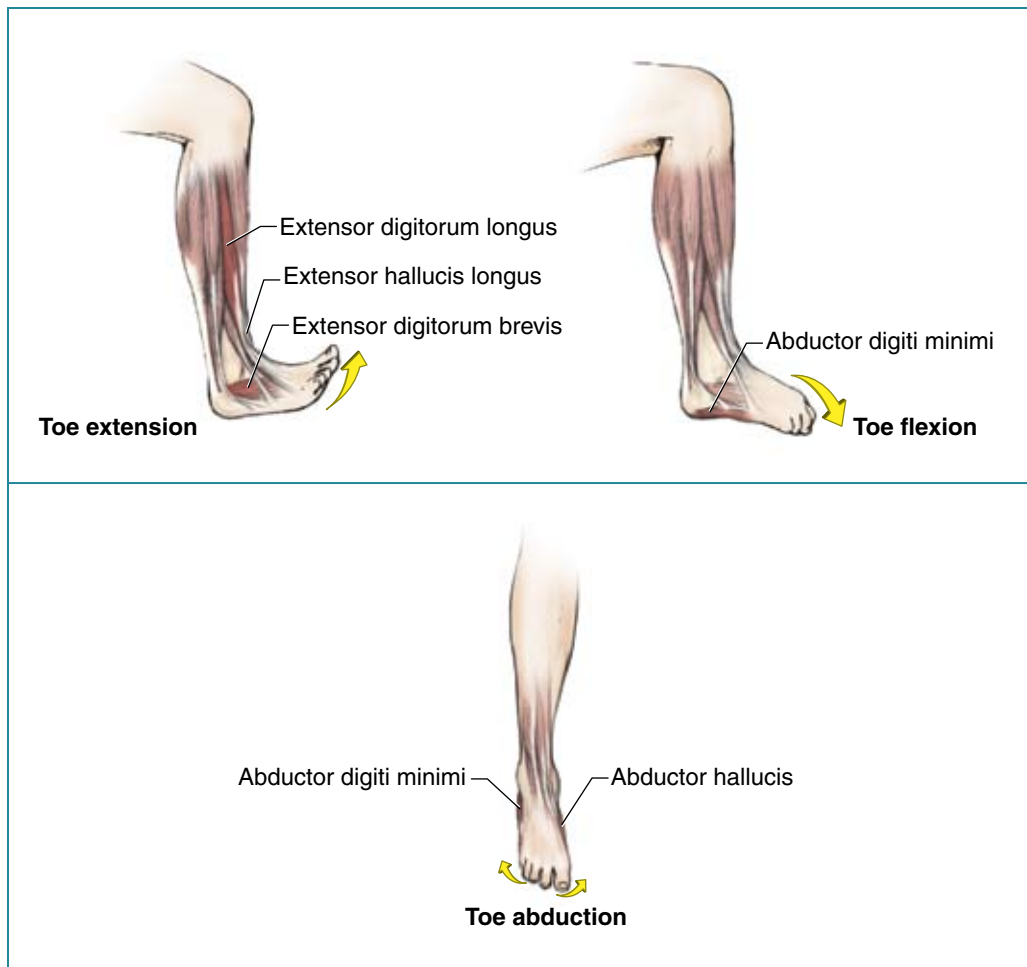
Commentary

It is more comfortable to do this stretch while wearing shoes. This is an excellent stretch for the flexor digitorum longus, medial soleus, and abductor hallucis muscles, which are located at the medial side of the lower leg and foot. Take extra care when placing the foot in an everted position, and make sure to progress slowly through this stretching position. After the right heel reaches the floor or the lowest point, bending the right knee slightly can increase the stretch. This reduces the stretch on hamstring muscles, but it increases the stretch on the calf muscles.

Foot and Calf Muscle Movements

The stretches in this chapter are excellent overall stretches; however, not all of these stretches may be completely suited to each person's needs. The muscles involved in the various calf, ankle, and foot movements appear in the following table. To stretch specific muscles, the stretch must involve one or more movements in the opposite direction of the desired muscle's movements. For example, if you want to stretch the left flexor digitorum longus, you could perform a movement that involves dorsiflexion and eversion of the left ankle and toe extension of the left foot. When a muscle has a high level of stiffness, you should use fewer simultaneous opposite movements (for example, to stretch a very tight flexor digitorum longus, you could start by doing only toe extension). As a muscle becomes loose, you can incorporate more simultaneous opposite movements.





LOWER LEG

Muscle	Dorsi- flexion	Plantar flexion	Foot inversion	Foot eversion	Toe extension	Toe flexion
Anterior tibialis	✓		✓			
Extensor digitorum longus	✓			✓	✓	
Extensor hallucis longus	✓		✓		✓	
Peroneus tertius	✓			✓		
Peroneus longus		✓		✓		
Peroneus brevis		✓		✓		
Gastrocnemius		✓				
Soleus		✓				
Plantaris		✓				
Posterior tibialis		✓	✓			
Flexor digitorum longus		✓	✓			✓
Flexor hallucis longus		✓	✓			✓

FOOT

Muscle	Toe extension	Toe flexion	Toe abduction
Flexor digitorum brevis		✓	
Quadratus plantae		✓	
Flexor digiti minimi brevis		✓	
Abductor hallucis		✓	✓
Abductor digiti minimi		✓	✓
Adductor hallucis		✓	
Flexor hallucis brevis		✓	
Extensor digitorum brevis	✓		
Extensor hallucis brevis	✓		
Lumbricales	✓	✓	
Plantar interosseus		✓	
Dorsal interosseus	✓		

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